Predictive Modeling Using Machine Learning

1. Introduction

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions. The objective of this project is to build and evaluate machine learning classification models using the Iris flower dataset.

2. Objective

The main objectives of this project are:

- To explore and understand a dataset.
- To prepare features and target labels for machine learning.
- To train classification models.
- To evaluate and compare model performance.
- To visualize model results.

3. Dataset Description

The Iris dataset contains 150 flower samples belonging to three species: Setosa, Versicolor, and Virginica.

The dataset contains four numerical features:

1. Sepal length
2. Sepal width
3. Petal length
4. Petal width

The target variable represents the species of each flower.

4. Data Preparation

The dataset was loaded and examined using Python. The features and target labels were separated before training.

The dataset was divided into:

- Training data: 120 samples (80%)
- Testing data: 30 samples (20%)

Stratified splitting was used to preserve the species distribution in the training and testing sets.

5. Machine Learning Models

Two classification algorithms were implemented:

Decision Tree Classifier

A Decision Tree learns decision rules from training data to classify flower species.

Random Forest Classifier

A Random Forest combines multiple decision trees to make predictions. Combining trees can improve generalization, although performance depends on the dataset and model settings.

6. Model Evaluation

Accuracy was used to measure the percentage of correct predictions on the test dataset.

Model	Test Accuracy
Decision Tree	93.33%
Random Forest	90.00%

The Decision Tree correctly classified 28 out of 30 test samples. The Random Forest correctly classified 27 out of 30 test samples.

In this experiment, the Decision Tree achieved the higher test accuracy.

7. Visualizations

Two visualizations were created:

- Model Accuracy Comparison: compares the test accuracy of the Decision Tree and Random Forest.
- Decision Tree Confusion Matrix: displays the correct and incorrect classifications for each Iris species.

These visualizations help explain and compare the model results.

8. Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

9. Conclusion

This project demonstrated the basic machine learning workflow, including dataset exploration, data splitting, model training, prediction, evaluation, and visualization.

The Decision Tree achieved 93.33% test accuracy, while the Random Forest achieved 90.00%. Therefore, the Decision Tree performed better on this particular test split.

The results are based on a single train-test split. Further evaluation using cross-validation and additional performance metrics could provide a more reliable comparison.

10. Future Improvements

- Evaluate models using cross-validation.
- Compare additional classification algorithms.
- Examine precision, recall, and F1-score.
- Tune model hyperparameters.
- Test models on additional data.